SAABIQ U A

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate at College of Engineering Guindy, Anna University, with hands-on experience in RISC-V CPU design, RTL development, functional verification, cache design, and digital ASIC implementation. Experienced in Verilog/SystemVerilog and C, with exposure to computer architecture and processor design. Familiar with synthesis, timing analysis, and RTL-to-GDSII implementation using Yosys, OpenLane, OpenROAD, and SKY130A.

TECHNICAL SKILLS

Programming: C,C++

HDL & Design: Verilog HDL, SystemVerilog, RISC-V Assembly, RTL Design

Verification & Simulation: Icarus Verilog, GTKWave, Vivado, SystemVerilog Testbenches

ASIC Design & Physical Implementation: Yosys, OpenLane, OpenROAD, Magic, SKY130A

Systems & Tools: Linux, Git, GitHub, MATLAB, LTspice

PROJECTS

RV32I Processor Core | Team Project (Ongoing) **Jul 2026 -- Present**

- Designed the RTL architecture, datapath, and control logic of a 32-bit single-cycle RISC-V (RV32I) processor in SystemVerilog, integrating the ALU, register file, and instruction/data memory access logic.
- Developed SystemVerilog testbenches and used Vivado simulation to verify supported instruction execution, branching, and memory operations.
- Completed the single-cycle processor baseline; hardware multiplier integration and pipelining are planned next phases, with the multiplier being developed by a teammate.

Radix-4 Modified Booth Multiplier with Carry Lookahead Adder (CLA) (GitHub) **Jan -- Apr 2026**

- Designed and functionally verified an 8-bit signed Radix-4 Modified Booth multiplier with partial-product generation, carry-save reduction, and a 16-bit Carry Lookahead Adder (CLA) using Verilog HDL.
- Performed RTL synthesis and timing analysis, analyzing resource utilization and critical-path timing.
- Completed RTL-to-GDSII ASIC implementation using OpenLane/SKY130A, targeting 70% core utilization and achieving 6.76 ns post-route combinational-path delay with zero DRC/LVS errors.

Asynchronous FIFO Tapeout (GitHub) **Jan -- Sep 2025**

Tiny Tapeout Sky25A Shuttle | CEG Fabless Initiative (Alumni-Student Forum)

- Designed and verified a parameterized asynchronous FIFO using Gray-coded read/write pointers and two-stage synchronizers for reliable clock-domain crossing between independent clock domains.
- Developed Verilog RTL and simulation testbenches to validate FIFO full/empty detection, read/write operations, reset behavior, and CDC functionality across asynchronous clocks.
- Completed RTL-to-GDSII implementation using OpenLane and SKY130A 130 nm, including synthesis, floorplanning, placement, routing, physical verification, and generation of the final GDSII layout.

EXPERIENCE

RISC-V Design Intern **May -- Aug 2025**

CEG Fabless, Anna University, in collaboration with Vyoma Systems and SHAKTI Processor Group, IIT Madras

- Worked with RISC-V architecture and ISA, developing and testing assembly programs for arithmetic, control-flow, and memory operations using the UptickPro simulation framework.
- Implemented L1 I-Cache and D-Cache replacement-policy RTL using FIFO and LFU policies and verified functionality against SHAKTI processor benchmarks.
- Analyzed processor/cache behavior through simulation and debugging, gaining hands-on experience in RISC-V processor design and RTL verification.

EDUCATION

College of Engineering Guindy, Anna University, Chennai, Tamil Nadu **Sep 2023 -- Present**

B.E. Electronics and Communication Engineering

CGPA: 8.54

Relevant Coursework: Digital Electronics, Digital IC Design, Computer Architecture, Verilog HDL, Static Timing Analysis (STA)

CERTIFICATIONS & ACHIEVEMENTS

- Digital IC Design - **NPTEL** (*2025*)
- Embedded Systems Design - **NPTEL** (*2025*)
- VLSI Design Flow: RTL-GDS - **NPTEL** (*2025*)
- Reliance Foundation Undergraduate Scholarship (*2023*)

LANGUAGES & INTERESTS

Languages: English, Tamil

Interests: CPU Design, RTL Design, ASIC Design, RISC-V

LEADERSHIP & POSITIONS OF RESPONSIBILITY

- Head of Coverage | PIXELS, CEG **Jul 2026 -- Present**
- Head of Editing | PIXELS, CEG **Jul 2025 -- Jul 2026**
- Junior Head of Editing | PIXELS, CEG **May 2024 -- Jul 2025**
- Webmaster | IEEE EDS, CEG **Mar 2025 -- Dec 2025**
- Department Representative | DECE, CEG **Aug 2025 -- Jul 2026**